

Foundations Notebook

Your Name

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Chapter 1

Mathematical Foundations

1.1 Calculus

1.1.1 Vector Calculus

Flux

1.2 Linear Algebra

1.2.1 Vector Spaces

Definition 1.2.1. A vector space V over a field $\mathbb{F}$ is ...

1.2.2 Why This Matters

Linear algebra underpins signal spaces, optimization, and estimation theory.

Chapter 2

Electrical Engineering Foundations

2.1 Electro Magnetism

2.1.1 coulombs Law

2.1.2 Maxwells Equations

$$\nabla \cdot \mathbf{D} = \rho, \quad (2.1)$$

$$\nabla \cdot \mathbf{B} = 0, \quad (2.2)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad (2.3)$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}, \quad (2.4)$$

where $\mathbf{E}$ is the electric field intensity in the units of V m^{-1} , $\mathbf{H}$ is the magnetic field intensity in the units of A m^{-1} , $\mathbf{D}$ is the electric flux (see section 1.1.1) densities in the units of C m^{-2} , and $\mathbf{B}$ is the magnetic flux density in the units of Wb m^{-2} or a Tesla(T), $\mathbf{D}$ is the electric displacement and (B) is magnetic induction.

2.1.3 Electric Field

This type of electric field is based static charge, and can be thought of like gravity such that the force being exerted is all around if in a 3 d manner that diminishes with the greater range from the source.

$$F = \frac{1}{4\pi\epsilon} \left(\frac{Q_1 Q_2}{r^2} \right) \quad (2.5)$$

2.2 Radar

Radar: Radio Detection And Ranging. Is simply the transmission and receiving of Radio waves. From these radio waves it is possible to determine many things about the incident target.

Simple Model

A transmitter is to be thought as a isotropic radiator with the propogation pattern described by equation 2.6.

$$P_d = \frac{P_t}{4\pi r^2} \quad (2.6)$$

Where P_d is the power desnity at some radius, P_t is the power transmitted and r is the radius.

$$R = \frac{c\delta t}{2} \quad (2.7)$$

2.2.1 SAR

2.3 Signals and Systems

2.3.1 Adaptive Filtering

What is an Adaptive filter

An adaptive filter uses a Wiener Filter structure but iterates it and adjusts it such that the error is reduced and it begins to approach the ideal filter.

$$h(n+1) = h(n) + \Delta h(n) \quad (2.8)$$

$$\Delta h(n) = -\gamma \left(\frac{d\sigma_{e_{\min}}^2}{dh} \right) \quad (2.9)$$

Applications

Using the ideas of Wiener Filters and Error Surfaces they can be applied in the ideas of signal recovery in a noisy channel and noise cancellation.

Noise cancellation Given a the noisy signal and correlated noise we are able to filter the correlated noise such that it matches the noise of the signal and thus allows us to cancel it out leaving only the signal in its place.

2.3.2 Sampling

2.3.3 LTI Systems

An LTI system satisfies:

- Linearity
- Time invariance

2.3.4 Idea

- Optical PCB's

Chapter 3

My Human Experience

3.1 Ideals

1. Give a shit
2. Know my shit
3. Don't be a shit
4. Get shit done

3.1.1 My Confessions

1. I struggle to accept my own failures or deficiencies particularly in the area of fitness and academics. When I feel like I am deficient I will push my insecurities out onto other people such that they are they are attacking me. By doing this I avoid facing my own insecurities and am able to make myself the victim.
2. I do not like giving defiant answers because I do not like to be wrong. By hiding in a grey area I can never be wrong.
3. I feel like I am drowning and everyone thinks I am lazy and not doing enough.
4. I am scared to leave academia, I feel like it is the only thing I am decent at.
5. Much of my logic and reasoning revolves around maintaining a degree of control, when that control is threatened I get very uncomfortable

and at times hostile/defensive. Attempting to address this feels incredibly overwhelming as it feels so deeply rooted into who I am.

3.2 Military Leadership

3.2.1 Objective

As a military officer my job is to manage the application of violence

3.2.2 Culture

Between branches the ideals revolve around the unit of action. In the case of the navy the unit of action is the ship, the army is the squad, and the airforce the plane. The ability for the unit of action to execute its mission is what drives the culture. In the case of the navy, for a ship to execute its mission the entire ship crew and all must act as a unified machine. Little deviation can be accepted as to ensure proper function of the ship. Hence the navy has a culture of detail driven instruction and execution. While the army being based around the squad needs to rely on the flexibility and innovation of the individuals in the squad to maintain effectiveness. Thus the army may focus more on the presentation and management of an officer as opposed to the navy's greater focus on technical ability.

3.2.3 After Action Reports

International Travel - France

Overall the trip was a success, Some key lessons are listed below

- 1.
- 2.